

Compute and fetch in sparse attention

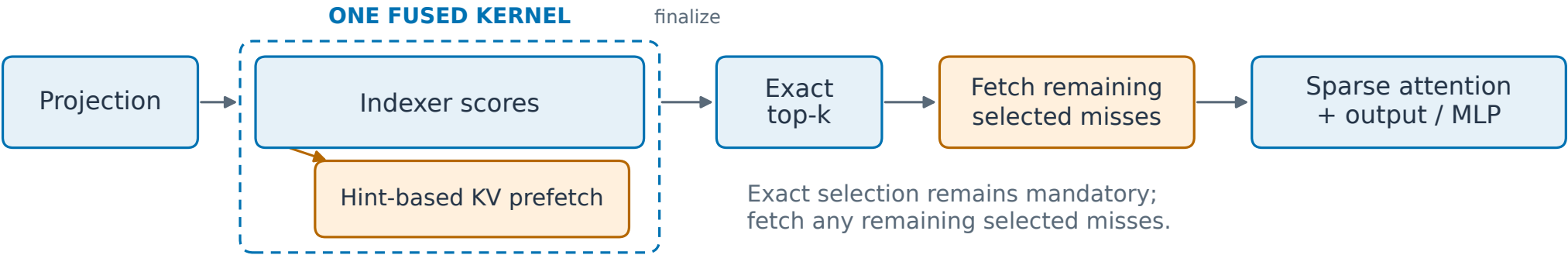
SCHEMATIC · NOT TO SCALE

DeepSeek C10 candidate execution · history KV misses · $H \leq P$



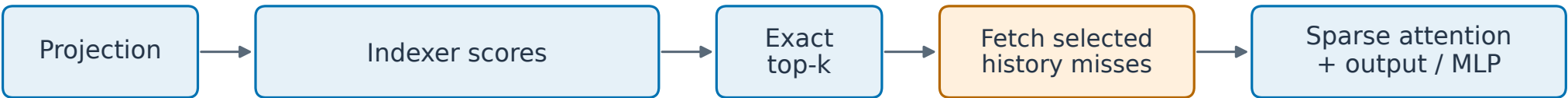
ECHO

In-layer fusion
SM compute + host reads



serial_sparse

Select first
then fetch exact misses

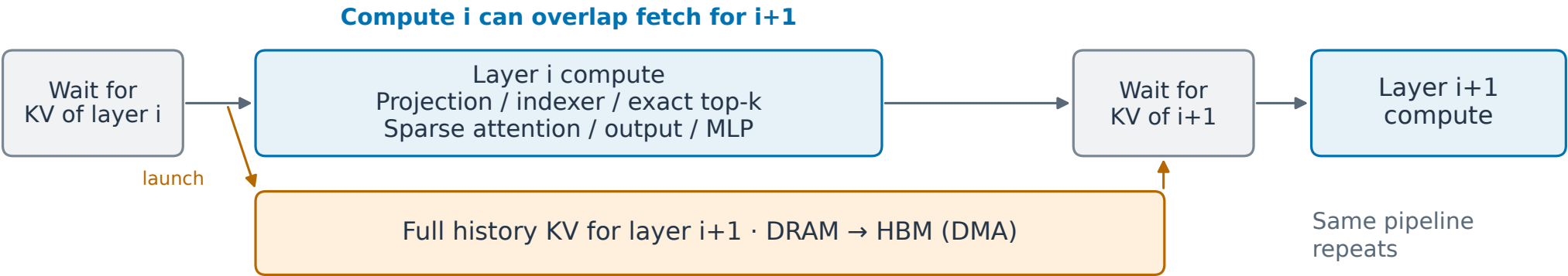


SM-based sparse fetch: attention starts after the selected KV is ready.

dense_prefetch

Cross-layer DMA

Independent
copy stream



History KV moves from CPU DRAM to HBM; candidate KV stays on the GPU.

Widths are not timings. Resident history can bypass fetch. Cache bookkeeping and pipeline startup/drain are simplified.